HDSI SCHOLARSHIP PROGRAM

The HDSI Undergraduate Scholarship Program is changing for 2026-7!

HDSI SCHOLARSHIP PROGRAM

- Before: form your own group; find your own mentor; research, design, and implement your own year-long, end-to-end data science project from scratch
- Now: we have designed seven project ideas with mentors, working with a real-world non-profit organization or public agency to help their mission:
  - Substance use intervention (Data Science Alliance w/ San Diego Rescue Mission)
  - Food bank distribution (Data Science Alliance w/ Feeding San Diego)
  - Disability services gaps (Data Science Alliance, for multiple govt and non-profit orgs)
  - Modeling homelessness: #1 economic; #2 agent-based (DSA w/ Lucky Duck Foundation)
  - UCSD Library: #1 trends in research/publishing; #2 AI/LLMs for navigating library resources
- You apply individually, then mentors run their own selection process and form teams of ~4

UC San Diego

Why Participate?

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- Gain experience with a real-world, live, messy, complex data problem
- Work with a mentor who is close to a specific applied domain
- Learn how organizations design, implement, and deliver a DS project
- Looks good on your resume, gives you something to brag about
- Mentors generally are happy to write letters of recommendation (but we can't promise for them)
- $2,250 stipend over the entire academic year, as SFS tuition+fee refund
  - Sent as refund (first applied to outstanding fees) 1/3rd each quarter
  - This is not a job, but this is about 4 hours/wk * 30 weeks at $18.75/hr

Eligibility

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- UCSD DSC majors or minors who will have passed DSC 80 by the end of Spring 2026
- **<u>You cannot</u>** apply if you have taken Senior Capstone or will take it anytime in 2026-7.
- Given this, we expect most eligible students to be applying at the end of their third year
- Must be enrolled in at least 12 units per quarter during the scholarship period (Fall, Winter, and Spring). If you graduate or drop below 12 units, you will not be awarded the stipend for that quarter.
- Must not have previously received this scholarship before

Application Process & Timeline

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- Individual applications will open May 1st, due May 31st (not yet open)
    - Can apply to at most two projects (not orgs)
    - 2 page resume with skills and past project experience
    - For each project, statement of purpose (2000 char max)
- HDSI screens for eligibility, sends applications to partner orgs. Mentors run their own selection and group formation process.
    - Mentors from partner orgs may or may not interview, etc.
- Acceptance notifications should be by end of July 2026
- Project starts Week 1 Fall 2026, ends Week 10 Spring 2027
    - We expect Fall will be more exploratory, where teams will co-design a project w/ the mentor, then implement it in Winter and Spring
    - Team must present at an in-person showcase late Spring '27

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Project #1: Drug Use Trajectory & Substance Intervention Timing Analysis

Organizational Partner: Data Science Alliance / San Diego Rescue Mission

Project Description: Join the Data Science Alliance (DSA) in building a predictive model that identifies when school staff should intervene with students at risk of program dropout due to substance use patterns.

Responsibilities:

- Cleaning and structuring drug test records and enrollment timelines
- Supporting feature engineering for trajectory and sequencing variables
- Assisting with survival analysis and dropout model development
- Helping build and QA dashboard visualizations (heatmaps, timeline charts)

Skills required:

- Proficiency in Python (pandas, numpy, matplotlib/seaborn)
- Comfort working with messy, real-world datasets
- Ability to document and communicate methodology clearly (written and verbal)
- Familiarity with version control (Git/GitHub)
- Self-direction and ability to make progress with partial information

Nice to have but not required:

- Experience with survival analysis or time-series classification
- Familiarity with logistic regression, decision trees, or sequence modeling
- Prior work with health, social services, or behavioral data
- Interest in data ethics and responsible use of sensitive client records

What You Will Gain: Hands-on experience with longitudinal behavioral data and exposure to survival analysis and predictive modeling in a real social services context.

Point of contact(s)

- Dr. Adir Mancebo Jr., DSA Data Science Manager, adir@datasciencealliance.org
- Shay Samat, DSA Data Scientist, shay@datasciencealliance.org

Full project descriptions:

stu.lu/hdsiusp2026

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Project #2: Economic Modeling to Predict Homeless System Inflow

Organizational Partner: Data Science Alliance / Lucky Duck Foundation

Project Description: Join the Data Science Alliance (DSA) in replacing a static inflow assumption in San Diego's homeless system simulation with a dynamic, data-driven econometric model tied to real economic indicators.

Responsibilities:

- Sourcing and wrangling economic indicator datasets (unemployment, housing costs, eviction trends)
- Running regression and ML model experiments for inflow prediction
- Documenting model selection rationale and feature importance results
- Integrating the finalized model into the existing discrete-event simulation

Skills required:

- Proficiency in Python (pandas, numpy, matplotlib/seaborn)
- Comfort working with messy, real-world datasets
- Ability to document and communicate methodology clearly (written and verbal)
- Familiarity with version control (Git/GitHub)
- Self-direction and ability to make progress with partial information

Nice to have but not required:

- Experience with econometrics, time-series modeling, or regression with lagged variables
- Familiarity with publicly available economic datasets (BLS, HUD, Census)
- Prior exposure to simulation modeling or system dynamics is a plus • Interest in housing policy

What You Will Gain: Practical experience with econometric modeling, feature selection, and simulation integration on a civic impact project.

Point of contact(s)

- Dr. Adir Mancebo Jr., DSA Data Science Manager, adir@datasciencealliance.org
- Shay Samat, DSA Data Scientist, shay@datasciencealliance.org

Full project descriptions: stu.lu/hdsiusp2026

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Project #3: Agent-Based Heterogeneity for the Homeless System Simulation

Organizational Partner: Data Science Alliance / Lucky Duck Foundation

Project Description: Join the Data Science Alliance (DSA) in extending San Diego's homeless system simulation by replacing its homogeneous population assumption with heterogeneous agents whose attributes — age, disability, veteran status — shape how they move through the system.

Responsibilities:

- Reviewing existing discrete-event simulation architecture and identifying extension points
- Researching and implementing agent attribute schemas and transition probability logic
- Running and validating hybrid DES-ABM model outputs against historical data
- Documenting modeling decisions and performance comparisons

Skills required:

- Proficiency in Python (pandas, numpy, matplotlib/seaborn)
- Comfort working with messy, real-world datasets
- Ability to document and communicate methodology clearly (written and verbal)
- Familiarity with version control (Git/GitHub)
- Self-direction and ability to make progress with partial information

Nice to have but not required:

- Familiarity with discrete-event simulation (DES) or agent-based modeling (ABM) frameworks such as SimPy
- Experience working with homeless services data or other social services datasets
- Background in probabilistic modeling, stochastic processes, or survival analysis
- Knowledge of San Diego's Continuum of Care system

What You Will Gain: Deep experience with hybrid simulation methodologies (DES + ABM), and exposure to systems modeling for social policy.

Point of contact(s)

- Dr. Adir Mancebo Jr., DSA Data Science Manager, adir@datasciencealliance.org
- Shay Samat, DSA Data Scientist, shay@datasciencealliance.org

Full project descriptions:

stu.lu/hdsiusp2026

Project #4: Nutritional Equity Analysis of Food Bank Distribution

Organizational Partner: Data Science Alliance / Feeding San Diego

Project Description: Join the Data Science Alliance (DSA) in building a nutritional equity measurement framework for food bank distribution, translating Feeding San Diego's weight-based distribution records into ZIP-level nutritional impact scores across their 200+ agency network.

Responsibilities:

- Mapping food category distribution data to nutritional value estimates
- Building and validating the ZIP-level Nutritional Equity Index (NEI) scoring model
- Conducting panel analysis across the 2021–2025 dataset to identify trends and risk
- Developing the food type reallocation optimization under supply and routing constraints

Skills required:

- Proficiency in Python (pandas, numpy, matplotlib/seaborn)
- Comfort working with messy, real-world datasets
- Ability to document and communicate methodology clearly (written and verbal)
- Familiarity with version control (Git/GitHub)
- Self-direction and ability to make progress with partial information

Nice to have but not required:

- Experience with spatial data analysis (GeoPandas, or similar)
- Familiarity with USDA nutritional databases
- Panel data or longitudinal analysis experience
- Interest in food systems, nutrition equity, or public health

What You Will Gain: Real-world experience with food systems data, spatial equity analysis, and optimization modeling.

Point of contact(s)

- Dr. Adir Mancebo Jr., DSA Data Science Manager, adir@datasciencealliance.org
- Shay Samat, DSA Data Scientist, shay@datasciencealliance.org

Full project descriptions:

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Project #5: Identifying Disability Services Gaps and Accessibility Barriers

Organizational Partner: Data Science Alliance

Project Description: Join the Data Science Alliance (DSA) in building a structural accessibility analysis of disability services in San Diego County — measuring not just where services exist on a map, but whether they're actually reachable by the people who need them most.

Responsibilities:

- Building a unified provider inventory across regional centers, paratransit, supported employment, and other service types
- Conducting mode-realistic travel time modeling and overlaying disability prevalence data
- Computing and validating the census tract-level Disability Services Accessibility Index (DSAI)
- Researching and mapping funding streams against DSAI gap findings

Skills required:

- Proficiency in Python (pandas, numpy, matplotlib/seaborn)
- Comfort working with messy, real-world datasets
- Ability to document and communicate methodology clearly (written and verbal)
- Familiarity with version control (Git/GitHub)
- Self-direction and ability to make progress with partial information

Nice to have but not required:

- Experience with spatial analysis and GIS tools (GeoPandas or OSMnx)
- Familiarity with ACS data or Census Bureau datasets
- Interest in disability rights, accessibility policy, or public health
- Prior work with travel time modeling is a strong plus

What You Will Gain: Hands-on experience with spatial analysis, accessibility modeling, and policy-facing data science in a County partnership context.

Point of contact(s)

- Dr. Adir Mancebo Jr., DSA Data Science Manager, adir@datasciencealliance.org
- Shay Samat, DSA Data Scientist, shay@datasciencealliance.org

Full project descriptions:

stu.lu/hdsiusp2026

Project #6: UCSD Library research and publishing trends

Organization: UC San Diego Library

Mentor: Stephanie Labou (Data Science Librarian)

Project description: The UC San Diego Library oversees a collections budget of over $9 million to fund access to books, journals, data, and other academic resources to a campus of over 45,000 students and 8,000 faculty and researchers. Having a better understanding of the research impact of the Library's collections – especially in regards to publication trends in peer-reviewed journals – will help inform collections management decisions in the coming fiscal years.

Guiding questions for this project include:

- In which journals do UC San Diego faculty publish most often and how do publishing trends vary by school/department?
- Which journals do UC San Diego faculty cite most often, and does the Library provide access to those journals?
- Where do UC San Diego researchers share their research data and other non-journal research outputs?
- Looking at abstracts of published research, what are emerging research areas popular on campus?

Skills expected: To answer these questions, students will need to retrieve data via API queries across multiple platforms (Dimensions, OpenAlex, and DataCite), use SQL to manage large tabular datasets in a relational database (Google BigQuery), and write and thoroughly document Python and/or R code to clean and wrangle disparate internal Library, campus, and UC-level datasets, which currently exist as lots of spreadsheets.

Nice to have, but not required: network analysis / citation network visualization; natural language processing; visualization platforms like Shiny, Streamlit, or Looker Studio; testing AI tools (Gemini, Notebook LM) for usability in data processing workflow.

Other details: Students will work with the Data Science Librarian, Stephanie Labou, who will provide existing proof-of-concept code and workflow to use as a starting point, as well as access to internal Library datasets. Expected project deliverables include a written report, a presentation to internal Library stakeholders, and an interactive dashboard for public-facing exploration of 10-15 years of UC San Diego researcher publishing trends. Students may also be featured in the UC San Diego Library newsletter.

Full project descriptions: stu.lu/hdsiusp2026

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Project #7: AI/LLM/chatbots for navigating library resources

Organization: UC San Diego Library

Mentor: Stephanie Labou (Data Science Librarian)

Project description: The Library has thousands of pages of documentation and guidance for navigating our resources - a website with multiple sections for everything from hours and online access to the A-Z database list and hundreds of course and research guides - yet students often struggle to find what they need quickly and easily. This project would investigate the feasibility of using existing online Library content to create a Library-specific chatbot or custom LLM.

Potential guiding framework for project: identify and describe the gap between current structure of resource documentation and expected input for LLM in order to achieve most robust answers; suggest structured prompts to incorporate existing documentation into existing LLM and/or develop a user guide for how to tailor existing materials to be LLM-ready; demonstrate 2-3 examples of LLMs customized to specific Library services. Students will work with the Data Science Librarian, Stephanie Labou, and other Library stakeholders as appropriate.

Skills expected: use of UC San Diego appropriate generative AI / LLM platforms (Notebook LM, Gemini, TritonGPT) for testing, plus Python code as needed. Methodology for a Library-specific chatbot/LLM (prompt engineering, fine-tuning, retrieval-augmented generation, etc.) will be up to the students.

Other details: Students will work with the Data Science Librarian, Stephanie Labou, who will provide existing proof-of-concept code and workflow to use as a starting point, as well as access to internal Library datasets. Expected deliverable is a written report detailing the current appropriateness of Library resources documentation as input/source information for potential LLM/chatbot options, plus proof-of-concept custom LLM/chatbot to demonstrate function in practice. Students will be expected to give a presentation of their results to an internal Library audience and may also be featured in the UC San Diego Library newsletter.

Full project descriptions:

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Application Form Will Ask

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- Applicant Information: Name(s), PID(s), UCSD Email(s), Major(s)/Minor(s), Planned Graduation Month/Year(s)
- Applicant resume (up to 2 pages max, PDF), including technical and project management skills, and past project experience (if any)
- Academic record / unofficial transcript (from WebReg, PDF)
- Hours per week you are generally able to commit (minimum 4)
- For each project (max 2), a statement of purpose (max 2000 characters including spaces, or ~300 words)

Statement of Purpose

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- One for each project you apply to, max 2000 chars (~300 words)
  - If you apply to both library or DSA homelessness projects, your two statements may overlap. Write them independently so each stands alone
  - Why are you a good fit for this organization and project?
  - What can you bring to this project & organization beyond what your resume and transcript shows?
- You can answer this in many ways, including but not limited to:
  - Past projects that show your skills: technical, management, collaboration, or other
  - Previous experiences that have shaped the way you would approach this project
  - What draws you to their mission and this project specifically
  - Connections you have to the work they do or the communities they serve
  - Or your views about the role of data science in society

On May 1st the application will open at:
stu.lu/hdsiusp2026apply

This will all be linked from the UCSD HDSI website –> Undergraduate –> Current Students –> Financial Opportunities (but is not yet updated)

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